Speed of light and sound

5a

The speed of waves varies depending on the medium (i.e. substrate, substance) the wave is travelling through. For light waves, denser mediums slow down the waves. For sound waves, denser mediums increase the speed. Sound waves must travel through a medium. A vibrating particle in this medium bumps into another particle, which then also starts to vibrate and so bumps into yet another particle. The kinetic particle theory can be used to explain the speed of sound in different mediums.

1 Explain how an iron bar conducts sound faster than air does.
(Hint: Think of the proximity of particles.)

2 Predict and explain whether cold air particles (molecules) will transmit sound at a higher speed than hot air particles. (Hint: Think of the speed of particles.)

Light waves are not dependent on the presence of a medium. Their electromagnetic vibrations can travel through a vacuum as well as a medium. However, when light waves spread through a medium, their energy is absorbed by the atoms of the medium and then given out again.

3 Explain why light travels more slowly in glass than it does in air.

4 Explain why air allows light to travel through it at nearly the same speed as in a vacuum?

5 Light travels at a speed of 300 000 000 m/s and sound at 333 m/s.

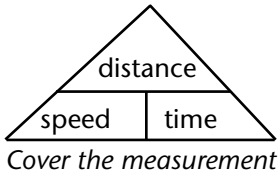
a Convert these speeds into km/s. (Hint: 1 m equals 1/1000 km.)

b Now convert these speeds into km/h. (Hint: 1 s equals 1/3600 h.)

6 Timekeepers at the finish line of a race are told to watch the smoke of the starting gun rather than listen for the sound. Explain.

7 A plane in the air looks further forward than it sounds. Explain.

8 Six km away fireworks are shot into the air. Calculate how soon you are able to see and hear them. (Hint: Recall the formula for speed)



Cover the measurement you want with your finger

9 During a thunderstorm you see a lightning flash and three seconds later you hear the thunder. How far away is the thunderstorm?

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Speed of light and sound

5b

10 While engaged in intergalactic space travel, a starship which is about 3 000 000 km away from you explodes. Calculate how quickly you would see and hear that explosion.

_____

_____

11 The speed of a wave is calculated by multiplying its wavelength by its frequency. State units for these three quantities:

speed: _____ wavelength: _____ frequency: _____

12 **a** Write the formula for the speed calculation:

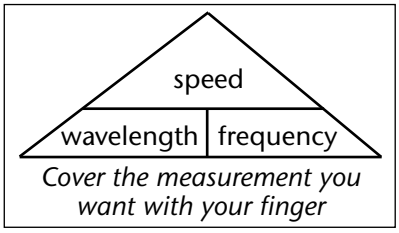
_____

b Adjust the formula: make wavelength the subject.

_____

c Adjust again: make frequency the subject.

_____



13 The following table shows information about musical notes. Complete the table using the speed formula or any of its adjustments. Show your working in the Calculation column.

	Frequency (Hz)	Speed at 20°C (m/s)	Calculation	Wavelength (m)
Middle C	256	342		
High C	512	342		
Low C	128	342		
Concert pitch (middle A)	440	342		
Lowest heard sound	20	342		
Highest heard sound	20 000	342		

14 The speed of light has been given the symbol c; it is an important number. c is also the speed of all electromagnetic radiation waves. State the speed of light in m/s and in km/s.

_____

15 The following table shows the frequencies and wavelengths of several electromagnet waves. Complete it by referring to the formulas in question 12.
(Hint: The speed of light is 300 000 000 m/s or 3×10^8 m/s.)

Wave type	Frequency (Hz)	Wavelength (m)	Calculation
Gamma rays	$3 . 10^{19}$		
Red light		$750 . 10^{-9}$	
Blue light	$6.7 . 10^{14}$		
Radio waves		10^2	

Note: To multiply two exponential numbers, add their exponents: e.g. $10^3 . 10^{-2} = 10^{3+(-2)} = 10^{3-2} = 10^1$ (i.e. 1000 . 0.01 = 10). To divide two exponential numbers, subtract the exponents: e.g. $10^5 / 10^3 = 10^{5-3} = 10^2$ (i.e. 100 000 / 1 000 = 100); $10^2 / 10^{-3} = 10^{2-(-3)} = 10^{2+3} = 10^5$. Numbers without exponents are treated in the usual way: e.g. $6 . 10^6 / 3 . 10^3 = 6 / 3 . 10^{6-3} = 2 . 10^3$.

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